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Unit “ 4 : Session “ 6 : SLO “ 1 | Parallel Computing Paradigms and Key Models

■ I. Fill in the Blanks

1. Parallel “ parallel computing allows multiple processes to run simultaneously.
2. Shared “ shared-memory architecture gives all processors one global memory space.

■ II. Multiple Choice

1. b) Shared Memory “ tightly coupled processors communicate through a common address space with low latency, so the shared-memory model suits them best.

■ III. Key Notes

- Shared memory: tightly coupled, communication via variables in common memory (e.g. multicore CPUs, OpenMP).
- Distributed memory: loosely coupled, each node owns private memory, communication via message passing (e.g. clusters, MPI).